Assignment 1

Due at 11:59pm on September 16.

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This assignment is to be submitted individually. Turn in this assignment as an HTML or PDF file to ELMS. Make sure to include the R Markdown or Quarto file that was used to generate it. You should include the questions in your solutions. You may use the qmd file of the assignment provided to insert your answers.

Git and GitHub

1) Provide the link to the GitHub repo that you used to practice git from Week 1. It should have:

- Your name on the README file.
- At least one commit with your name, with a description of what you did in that commit.

https://github.com/zzeng05/Zeng1-Liu2-a1.git

Reading Data

Download both the Angell.dta (Stata data format) dataset and the Angell.txt dataset from this website: https://stats.idre.ucla.edu/stata/examples/ara/applied-regression-analysis-by-fox-data-files/

2) Read in the .dta version and store in an object called `angell_stata`.

```r
library(haven)
angell_stata <- read_dta("/Users/zpzzz/Desktop/SURV727/Zeng1-Liu2-a1/angell.dta")
summary(angell_stata)
```

```
     city               morint          ethhet          geomob     
 Length:43          Min.   : 4.20   Min.   :10.60   Min.   :12.10  
 Class :character   1st Qu.: 8.70   1st Qu.:16.90   1st Qu.:19.45  
 Mode  :character   Median :11.10   Median :23.70   Median :25.90  
                    Mean   :11.20   Mean   :31.37   Mean   :27.60  
                    3rd Qu.:13.95   3rd Qu.:39.00   3rd Qu.:34.80  
                    Max.   :19.00   Max.   :84.50   Max.   :49.80  
    region         
 Length:43         
 Class :character  
 Mode  :character  
```

3) Read in the .txt version and store it in an object called `angell_txt`.

```
angell_txt <- read.table("/Users/zpzzz/Desktop/SURV727/Zeng1-Liu2-a1/angell.txt")
summary(angell_txt)
```

```
      V1                  V2              V3              V4       
 Length:43          Min.   : 4.20   Min.   :10.60   Min.   :12.10  
 Class :character   1st Qu.: 8.70   1st Qu.:16.90   1st Qu.:19.45  
 Mode  :character   Median :11.10   Median :23.70   Median :25.90  
                    Mean   :11.20   Mean   :31.37   Mean   :27.60  
                    3rd Qu.:13.95   3rd Qu.:39.00   3rd Qu.:34.80  
                    Max.   :19.00   Max.   :84.50   Max.   :49.80  
      V5           
 Length:43         
 Class :character  
 Mode  :character  
```

4) What are the differences between `angell_stata` and `angell_txt`? Are there differences in the classes of the individual columns?

```
names(angell_stata)
```

```
[1] "city"   "morint" "ethhet" "geomob" "region"
```

```r
names(angell_txt)
```

```
[1] "V1" "V2" "V3" "V4" "V5"
```

```r
sapply(angell_stata, class)
```

```
       city      morint      ethhet      geomob      region
"character"   "numeric"   "numeric"   "numeric" "character"
```

```r
sapply(angell_txt, class)
```

```
         V1          V2          V3          V4          V5
"character"   "numeric"   "numeric"   "numeric" "character"
```

```r
head(angell_txt)
```

```
          V1   V2   V3   V4 V5
1  Rochester 19.0 20.6 15.0  E
2   Syracuse 17.0 15.6 20.2  E
3  Worcester 16.4 22.1 13.6  E
4       Erie 16.2 14.0 14.8  E
5  Milwaukee 15.8 17.4 17.6 MW
6 Bridgeport 15.3 27.9 17.5  E
```

```r
head(angell_stata)
```

```
# A tibble: 6 x 5
  city       morint ethhet geomob region
  <chr>       <dbl>  <dbl>  <dbl> <chr>
1 Rochester    19     20.6   15   E
2 Syracuse     17     15.6   20.2 E
3 Worcester    16.4   22.1   13.6 E
4 Erie         16.2   14     14.8 E
5 Milwaukee    15.8   17.4   17.6 MW
6 Bridgeport   15.3   27.9   17.5 E
```

5) Make any updates necessary so that `angell_txt` is the same as `angell_stata`.

```r
## Rename columns
names(angell_txt) <- names(angell_stata)

## Align classes
angell_txt$city   <- as.character(angell_txt$city)
angell_txt$morint <- as.numeric(angell_txt$morint)
angell_txt$geomob <- as.numeric(angell_txt$geomob)
angell_txt$region <- factor(
  angell_txt$region,
  levels = levels(haven::as_factor(angell_stata$region))  # "E","MW","W","S"
)

## Cross check
print(names(angell_stata))
```

```
[1] "city"   "morint" "ethhet" "geomob" "region"
```

```r
print(names(angell_txt))
```

```
[1] "city"   "morint" "ethhet" "geomob" "region"
```

```r
print(head(angell_stata))
```

```
# A tibble: 6 x 5
  city       morint ethhet geomob region
  <chr>       <dbl>  <dbl>  <dbl> <chr>
1 Rochester    19     20.6   15   E
2 Syracuse     17     15.6   20.2 E
3 Worcester    16.4   22.1   13.6 E
4 Erie         16.2   14     14.8 E
5 Milwaukee    15.8   17.4   17.6 MW
6 Bridgeport   15.3   27.9   17.5 E
```

```r
print(head(angell_txt))
```

```
        city morint ethhet geomob region
1  Rochester   19.0   20.6   15.0      E
2   Syracuse   17.0   15.6   20.2      E
3  Worcester   16.4   22.1   13.6      E
```

```
4       Erie   16.2   14.0   14.8      E
5  Milwaukee   15.8   17.4   17.6     MW
6 Bridgeport   15.3   27.9   17.5      E
```

```r
print(all.equal(as.data.frame(angell_txt),
                as.data.frame(angell_stata),
                check.attributes = FALSE))
```

```
[1] "Component \"morint\": Mean relative difference: 2.311301e-08"
[2] "Component \"ethhet\": Mean relative difference: 2.475718e-08"
[3] "Component \"geomob\": Mean relative difference: 2.394829e-08"
[4] "Component \"region\": 'current' is not a factor"
```

6) Describe the Ethnic Heterogeneity variable. Use descriptive statistics such as mean, median, standard deviation, etc. How does it differ by region?

```r
library(psych)

psych::describe(angell_stata$ethhet)
```

```
   vars  n  mean    sd median trimmed   mad  min  max range skew kurtosis   se
X1    1 43 31.37 20.41   23.7   28.33 12.01 10.6 84.5  73.9 1.18      0.3 3.11
```

```r
psych::describeBy(angell_stata$ethhet, angell_stata$region, mat = TRUE)
```

```
    item group1 vars  n     mean        sd median  trimmed       mad  min  max
X11    1      E    1  9 23.48889 10.773398  22.10 23.48889  9.636900 10.6 45.8
X12    2     MW    1 14 21.67857  9.084914  19.25 21.09167  8.154301 10.7 39.7
X13    3      S    1 14 52.48571 21.440897  53.80 52.46667 27.353968 20.7 84.5
X14    4      W    1  6 16.55000  4.164012  16.15 16.55000  3.558239 12.3 23.9
    range         skew   kurtosis       se
X11  35.2  0.747188465 -0.5716749 3.591133
X12  29.0  0.734178773 -0.7125926 2.428045
X13  63.8 -0.001023661 -1.4947575 5.730321
X14  11.6  0.639974099 -1.1117407 1.699951
```

Describing Data

R comes also with many built-in datasets. The "MASS" package, for example, comes with the "Boston" dataset.

7) Install the "MASS" package, load the package. Then, load the Boston dataset.

```
if (!requireNamespace("MASS", quietly = TRUE)) install.packages("MASS") # for smooth renderi
library(MASS)
data("Boston")
```

8) What is the type of the Boston object?

```
typeof(Boston)
```

```
[1] "list"
```

9) What is the class of the Boston object?

```
class(Boston)
```

```
[1] "data.frame"
```

10) How many of the suburbs in the Boston data set bound the Charles river?

```
head(Boston)
```

```
     crim zn indus chas   nox    rm  age    dis rad tax ptratio  black lstat
1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98
2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14
3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03
4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94
5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90  5.33
6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
  medv
1 24.0
2 21.6
3 34.7
4 33.4
5 36.2
6 28.7
```

```r
sum(Boston$chas == 1)
```

```
[1] 35
```

11) Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each variable.

```r
head(Boston[order(-Boston$crim), ])
```

```
       crim zn indus chas   nox    rm   age    dis rad tax ptratio  black lstat
381 88.9762  0  18.1    0 0.671 6.968  91.9 1.4165  24 666    20.2 396.90 17.21
419 73.5341  0  18.1    0 0.679 5.957 100.0 1.8026  24 666    20.2  16.45 20.62
406 67.9208  0  18.1    0 0.693 5.683 100.0 1.4254  24 666    20.2 384.97 22.98
411 51.1358  0  18.1    0 0.597 5.757 100.0 1.4130  24 666    20.2   2.60 10.11
415 45.7461  0  18.1    0 0.693 4.519 100.0 1.6582  24 666    20.2  88.27 36.98
405 41.5292  0  18.1    0 0.693 5.531  85.4 1.6074  24 666    20.2 329.46 27.38
    medv
381 10.4
419  8.8
406  5.0
411 15.0
415  7.0
405  8.5
```

```r
head(Boston[order(-Boston$tax), ])
```

```
       crim zn indus chas   nox    rm  age    dis rad tax ptratio  black lstat
489 0.15086  0 27.74    0 0.609 5.454 92.7 1.8209   4 711    20.1 395.09 18.06
490 0.18337  0 27.74    0 0.609 5.414 98.3 1.7554   4 711    20.1 344.05 23.97
491 0.20746  0 27.74    0 0.609 5.093 98.0 1.8226   4 711    20.1 318.43 29.68
492 0.10574  0 27.74    0 0.609 5.983 98.8 1.8681   4 711    20.1 390.11 18.07
493 0.11132  0 27.74    0 0.609 5.983 83.5 2.1099   4 711    20.1 396.90 13.35
357 8.98296  0 18.10    1 0.770 6.212 97.4 2.1222  24 666    20.2 377.73 17.60
    medv
489 15.2
490  7.0
491  8.1
492 13.6
493 20.1
357 17.8
```

```r
head(Boston[order(-Boston$ptratio), ])
```

```
       crim zn indus chas   nox    rm  age     dis rad tax ptratio  black lstat
355 0.04301 80  1.91    0 0.413 5.663 21.9 10.5857   4 334    22.0 382.80  8.05
356 0.10659 80  1.91    0 0.413 5.936 19.5 10.5857   4 334    22.0 376.04  5.57
128 0.25915  0 21.89    0 0.624 5.693 96.0  1.7883   4 437    21.2 392.11 17.19
129 0.32543  0 21.89    0 0.624 6.431 98.8  1.8125   4 437    21.2 396.90 15.39
130 0.88125  0 21.89    0 0.624 5.637 94.7  1.9799   4 437    21.2 396.90 18.34
131 0.34006  0 21.89    0 0.624 6.458 98.9  2.1185   4 437    21.2 395.04 12.60
    medv
355 18.2
356 20.6
128 16.2
129 18.0
130 14.3
131 19.2
```

```r
summary(Boston[c("crim","tax","ptratio")])
```

```
      crim                tax           ptratio     
 Min.   : 0.00632   Min.   :187.0   Min.   :12.60  
 1st Qu.: 0.08205   1st Qu.:279.0   1st Qu.:17.40  
 Median : 0.25651   Median :330.0   Median :19.05  
 Mean   : 3.61352   Mean   :408.2   Mean   :18.46  
 3rd Qu.: 3.67708   3rd Qu.:666.0   3rd Qu.:20.20  
 Max.   :88.97620   Max.   :711.0   Max.   :22.00  
```

-

12) Describe the distribution of pupil-teacher ratio among the towns in this data set that have a per capita crime rate larger than 1. How does it differ from towns that have a per capita crime rate smaller than 1?

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following object is masked from 'package:MASS':

    select
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(ggplot2)
```

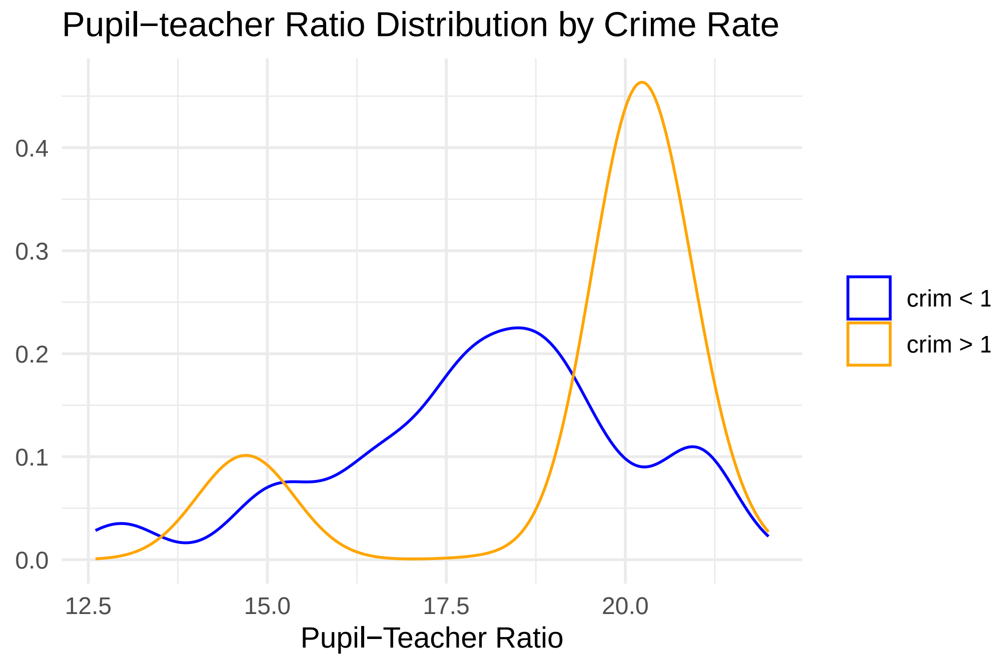
```
Attaching package: 'ggplot2'

The following objects are masked from 'package:psych':

    %+%, alpha
```

```r
df <- Boston %>%
  filter(crim != 1) %>%
  mutate(group = if_else(crim > 1, "crim > 1", "crim < 1"))

ggplot(df, aes(x = ptratio, color = group)) +
  geom_density() +
  labs(title = "Pupil-teacher Ratio Distribution by Crime Rate",
       x = "Pupil-Teacher Ratio", y = " ") +
  scale_color_manual(values = c("crim > 1" = "orange", "crim < 1" = "blue")) +
  theme_minimal() +
  labs(color=NULL)
```




```
by(df$ptratio, df$group, summary)
```

```
df$group: crim < 1
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  12.60   16.80   18.30   18.02   19.20   22.00 
------------------------------------------------------------
df$group: crim > 1
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  14.70   20.20   20.20   19.29   20.20   21.20 
```

Writing Functions

13) Write a function that calculates 95% confidence intervals for a point estimate. The function should be called `my_CI`. When called with `my_CI(2, 0.2)`, the function should print out "The 95% CI upper bound of point estimate 2 with standard error 0.2 is 2.392. The lower bound is 1.608."

```
my_CI <- function(pe, se, z = 1.96) {
  hi <- pe + z * se
  lo <- pe - z * se
  cat(paste0("The 95% CI upper bound of point estimate ", pe, " with standard error ", se, "
}
```

```r
# test
my_CI(2, 0.2)
```

```
The 95% CI upper bound of point estimate 2 with standard error 0.2 is 2.392. The lower bound
```

*Note: The function should take a point estimate and its standard error as arguments. You may use the formula for 95% CI: point estimate +/- 1.96*standard error.*

Hint: Pasting text in R can be done with: `paste()` *and* `paste0()`

14) Create a new function called `my_CI2` that does that same thing as the `my_CI` function but outputs a vector of length 2 with the lower and upper bound of the confidence interval instead of printing out the text. Use this to find the 95% confidence interval for a point estimate of 0 and standard error 0.4.

```r
my_CI2 <- function(pe, se, z = 1.96) {
  half <- z * se
  c(lower = pe - half, upper = pe + half)
}

# test
my_CI2(0, 0.4)
```

```
 lower  upper
-0.784  0.784
```

15) Update the `my_CI2` function to take any confidence level instead of only 95%. Call the new function `my_CI3`. You should add an argument to your function for confidence level.

```r
my_CI3 <- function(pe, se, conf.level = 0.95) {
  z <- qnorm(1 - (1 - conf.level)/2)
  half <- z * se
  c(lower = pe - half, upper = pe + half)
}

# test
 my_CI3(0, 0.4, 0.95)
```

```
     lower      upper
-0.7839856  0.7839856
```

```r
my_CI3(0, 0.4, 0.99)
```

```
    lower     upper 
-1.030332  1.030332 
```

Hint: Use the `qnorm` *function to find the appropriate z-value. For example, for a 95% confidence interval, using* `qnorm(0.975)` *gives approximately 1.96.*

16) Without hardcoding any numbers in the code, find a 99% confidence interval for Ethnic Heterogeneity in the Angell dataset. Find the standard error by dividing the standard deviation by the square root of the sample size.

```r
my_CI3(mean(angell_stata$ethhet),
       sd(angell_stata$ethhet) / sqrt(sum(angell_stata$ethhet))
       , conf.level = 0.99)
```

```
   lower    upper 
29.94061 32.80357 
```

17) Write a function that you can `apply` to the Angell dataset to get 95% confidence intervals. The function should take one argument: a vector. Use if-else statements to output NA and avoid error messages if the column in the data frame is not numeric or logical.

```r
CI95 <- function(v) {
  if (is.numeric(v) || is.logical(v)) {
    mu <- mean(v, na.rm = TRUE)
    se <- sd(v,   na.rm = TRUE) / sqrt(sum(!is.na(v)))
    z  <- qnorm(0.975)
    c(lower = mu - z * se, upper = mu + z * se)
  }
  else {
   cat("NA")
  }
}

# test
CI95(angell_stata$ethhet) # should output CI
```

```
   lower    upper 
25.27127 37.47292 
```

```r
CI95(angell_stata$city) # should output NA
```

```
NA
```